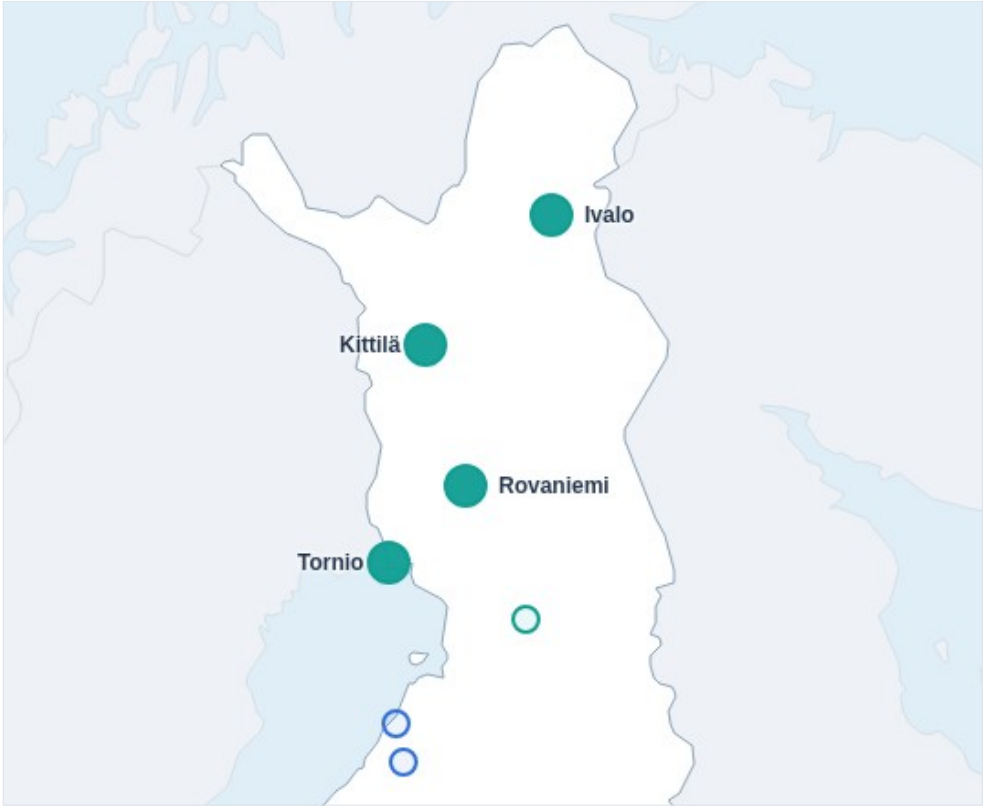
# In May 2005, Kittilä and Ivalo flooded together.

The snow had warned us two months earlier.

1 March snow flagged 2005 in advance. Kittilä and Ivalo then passed their flood levels in the same spring.



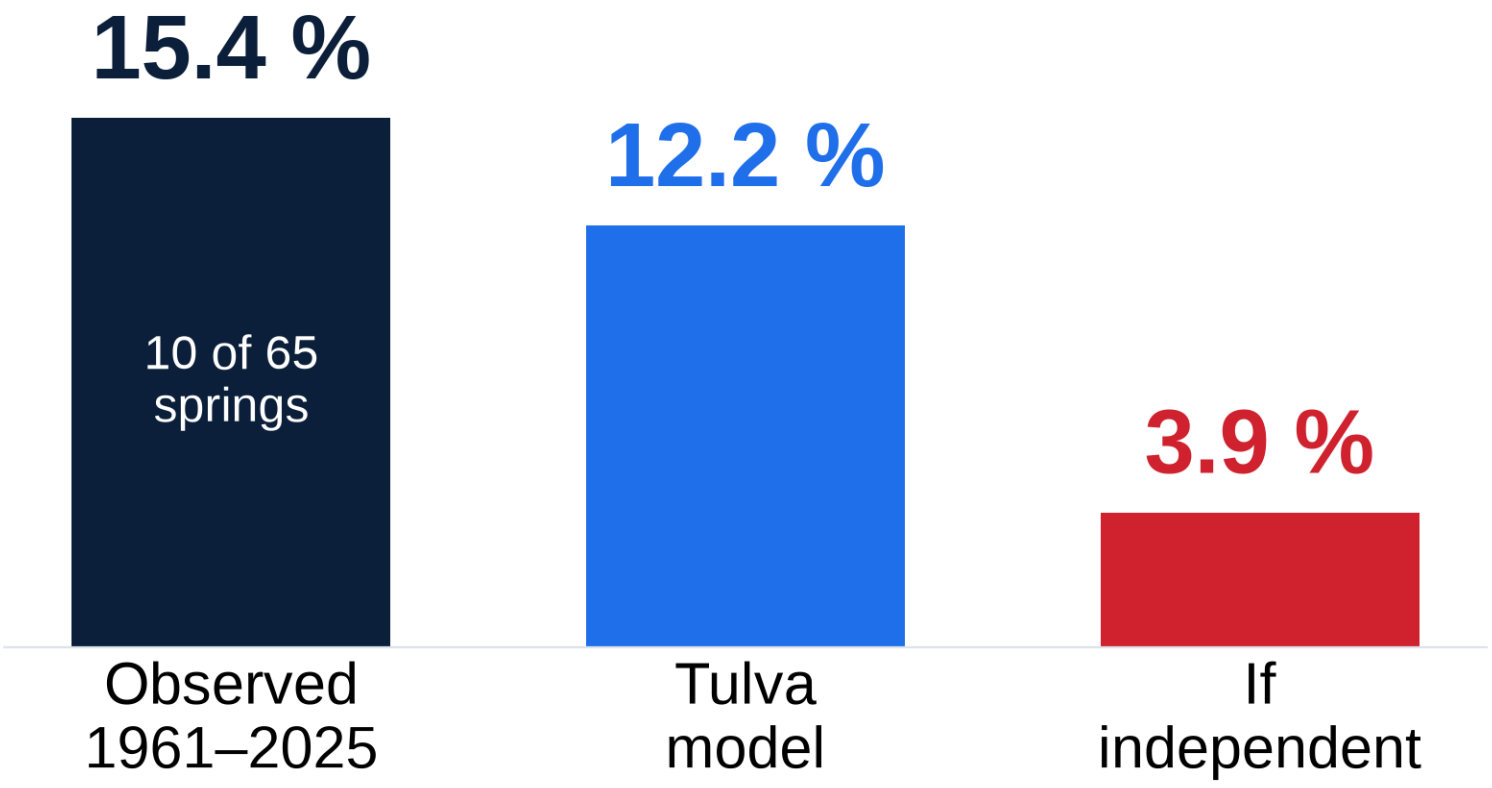

Tulva app, year replay: spring 2005

Source: SYKE discharge 1961–2025 and 1 March snow (SWE); Tulva back-test (out/backtest.json)

THE GAP

# Finland's rivers flood together, not one at a time

## How often do ≥4 of 14 river areas flood in the same spring?




Springs when ≥4 of the 14 river areas flooded (own 1-in-10 level)

**15.4 %**
observed, 1961–2025 (10 of 65 years)

**12.2 %**
Tulva joint model

**3.9 %**
if areas were independent

Home insurance pays only for floods rarer than 1-in-50.

Reinsurance renews 1 January, before the snow signal.

Source: SYKE 1961–2025 (+ SMHI Torne), Tulva model (exact); vesi.fi / OP (1-in-50 rule)

WHAT IT COSTS

# Priced as independent, a joint flood cover is ~5× too cheap

North cluster: pays €10M if ≥2 of 5 gauges pass their 1-in-50 level. Market price per year:

**€96k–134k**
if areas were independent

→

**€491k–687k**
Tulva joint risk (technical €265k)

**5.1×**
Tulva vs independence




Source: SYKE 1961–2025, Tulva model (exact; MC 1M agrees). Market = 2.5–3.5× expected payout (AM Best cat bonds, flood ILS)

THE PRODUCT

# Tulva Index: bought in winter, pays in spring

TULVA INDEX · NORTH CLUSTER

**Pays €10M** if ≥2 of 5 northern gauges pass their 1-in-50 level this spring

Independence

**€96k–134k**

market price per year

Tulva

**€491k–687k**

market price per year

After a snowy March

**€1.61M–2.26M**

market price per year

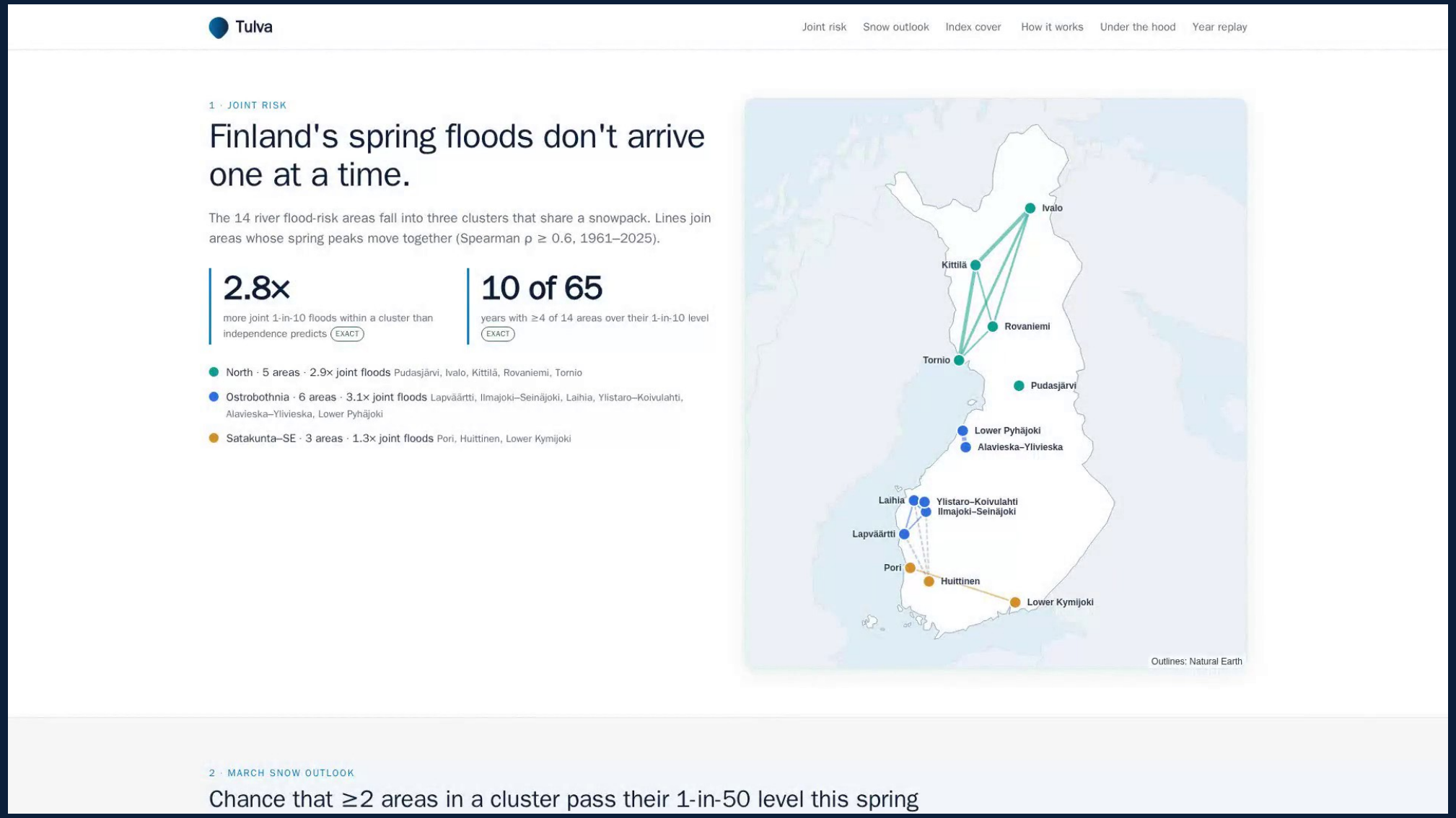
**Bought in winter**
before the snow signal exists

**Staged option**
50 % paid at 2 gauges, 100 % at 3

**Explainable trigger**
public SYKE gauges anyone can check

**11 of 19** payout years flagged in advance by the 1 March snow, 2005 included

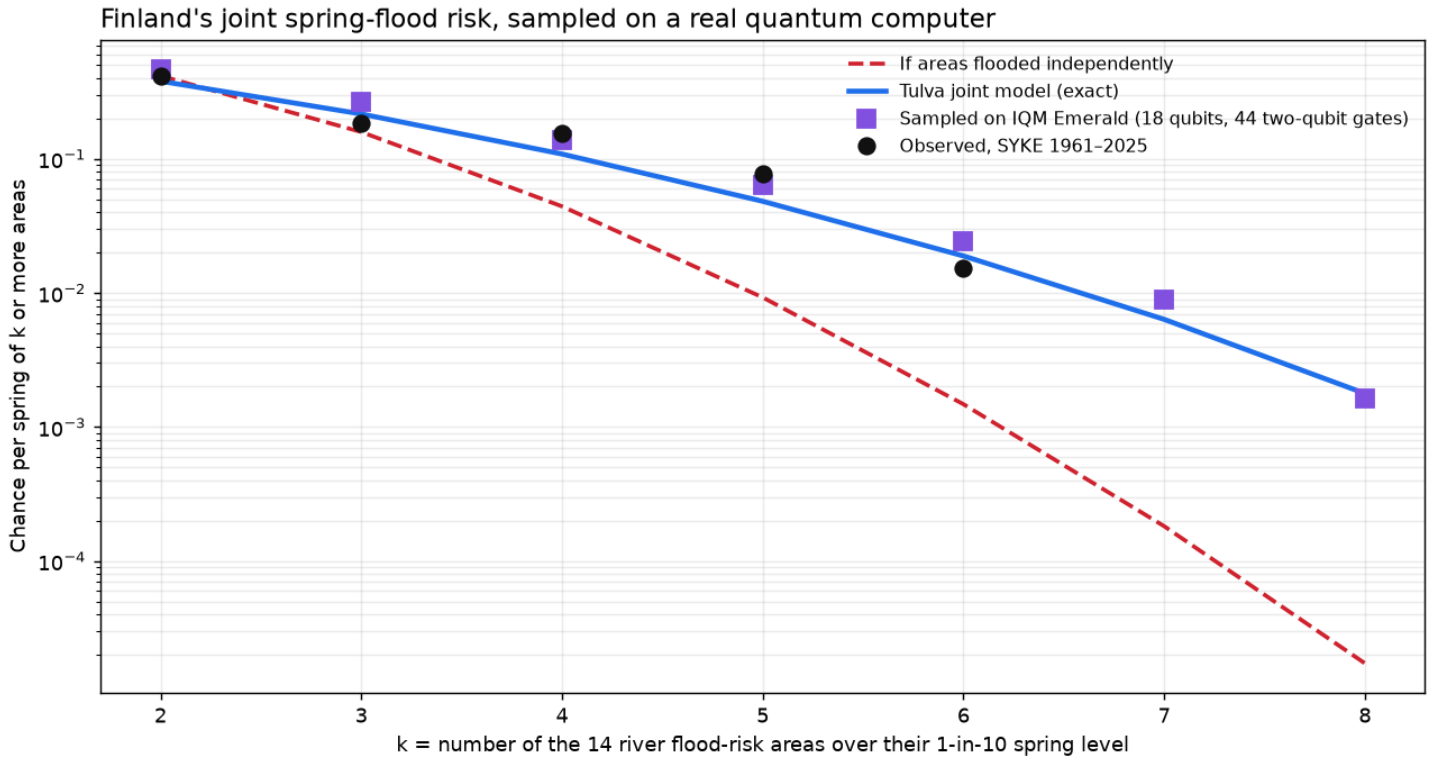
Source: Tulva model, exact; market 2.5–3.5× expected payout. Back-test 1961–2025, all clusters, ≥2 gauges over 1-in-10, in-sample




Live app: the joint-flood map → March snow outlook → index price → real quantum runs → 2005 replay

QUANTUM, IN PRACTICE

# Pricing = a joint-flood probability → quantum amplitude estimation




IQM Emerald via IQM Resonance, job 01a11cd4-da38-718b-80c9-7e259c6e4452, 8,000 shots, readout-corrected. Exact = two-level factor model fitted to SYKE spring peaks.

The chip reproduces the joint tail that independence misses.

ALL 14 AREAS ON IQM EMERALD

**18**
qubits

**44**
two-qubit gates

AMPLITUDE ESTIMATION, IBM & IQM

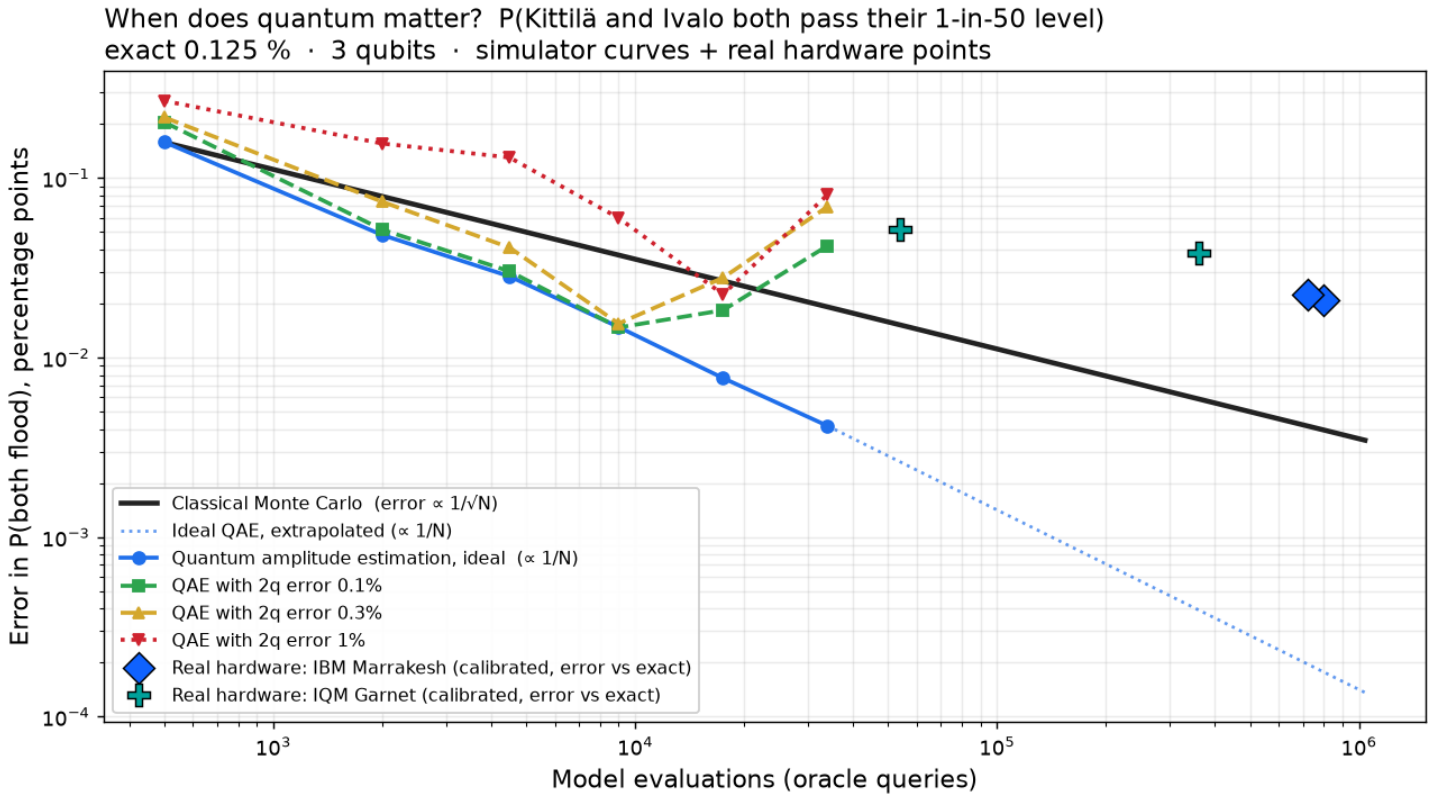
P(Kittilä and Ivalo both pass 1-in-50)

| | raw → calibrated |
|---|---|
| **Exact** | **0.125 %** |
| **IBM Marrakesh** | 0.262 → **0.148 %** |
| **IQM Garnet** | 0.368 → **0.163 %** |

**Snow signal on the chip:** **snowy vs low March 3.1× (exact 2.7×)**

IQM Emerald job 01a11cd4…, 8,000 shots; snow runs: Emerald, 10 copies, 240k shots each; IBM ibm_marrakesh (20 copies); IQM Garnet (4 copies); also native Qrisp on Garnet

# Classical Monte Carlo still wins today




Fitted to SYKE spring peaks 1961–2025. Noisy curves: depolarising noise + 1 % readout, square-lattice routing; simulator error = RMS over 200 repetitions.
Hardware points = |calibrated estimate − exact| of one run (mostly residual bias), at that run's total model evaluations.

## Quantum wins at scale, with:

**Fault-tolerant hardware**
~8k logical qubits for comparable pricing (Chakrabarti et al. 2021)

**Expensive models**
full hydrological simulation per scenario

**More work per price**
multi-peril, daily repricing

Ideal QAE error falls like 1/N, Monte Carlo like 1/√N.

Source: Tulva simulations (qae-sim) + IBM Marrakesh / IQM Garnet runs; Chakrabarti et al., Quantum 5, 463 (2021)

BUSINESS

# Who buys, why, and why now

WHO BUYS

## Finnish non-life insurers

For the flood they keep under reinsurance. Four groups write 91 % of the market. Then reinsurers and ILS investors.

HOW WE EARN

## Engine licence

Pricing and trigger engine, licensed per insurer or reinsurer. Structuring fee on placed covers.

WHY NOW

## 30 Jan 2027

Solvency II flood factor for Finland: 0 % correlation with SE/NO/DK (25 % for snowmelt rejected). ORSA needs climate scenarios.

SCALE

## Finland → Nordics → Central Europe

Same engine, any snowmelt or river region. Gävle 2021: SEK 1.9bn claims. Storm Boris 2024: €2.15bn insured.

**First step: a paid pilot with one insurer in autumn, ahead of the 1 January renewals.**

Sources: Finance Finland 2024; Del. Reg. (EU) 2026/269; EIOPA-BoS-24/462 §4.70; Finansinspektionen 2023 (Gävle); PERILS 2025 (Boris)

# Tulva prices Finland's spring floods as a joint risk, and runs it on real quantum hardware.

BUILT THIS WEEKEND

- SYKE data pipeline
- Joint flood model
- Index pricing
- QAE on IBM & IQM

**Toivo Kallio** · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack